

Karan Singh J

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Software Engineer | Java Full Stack Developer | Quantum ML Intern

Professional Summary

Software Engineer with 1.5+ years of professional experience in Java, Spring Boot, REST API development, SQL, and backend systems. Currently working as a Quantum Machine Learning Intern, gaining hands-on experience in quantum computing, machine learning, algorithm development, modelling, simulation, and research. Experienced in designing scalable applications, integrating APIs, and delivering end-to-end software solutions. Strong foundation in Object-Oriented Programming, database design, microservices architecture, and AI-driven systems.

Technical Skills

Languages: Java, SQL, Python, JavaScript
Backend: Spring Boot, Hibernate, Servlets, JDBC, FastAPI
Frontend: React, HTML5, CSS3, Tailwind CSS, Bootstrap
Databases: MySQL, Firebase
Tools & Platforms: Git, Docker
Core Concepts: OOP, RESTful APIs, Microservices

Experience Highlights

- 1.5+ years of professional experience across Software Development, Enterprise Operations, and Emerging Technologies.
- Currently working as a Quantum Machine Learning Intern focusing on quantum algorithms, modelling, simulation, and applied quantum technologies.
- Developed backend applications using Java, Spring Boot, REST APIs, and SQL.
- Built multiple full-stack, AI-driven, and machine learning applications from design to deployment.

Education

Rajarajeswari College of Engineering, Bengaluru

M.Tech - Computer Science and Engineering, CGPA: 8.61

2026

Visvesvaraya Technological University (VTU)

B.E. - Information Science and Engineering, CGPA: 6.96

2022

Professional Experience

Ada Lovelace Software Private Limited

Quantum Machine Learning Intern

March 2026 – Jun 2026

- Developed quantum computing and quantum machine learning applications using Qiskit, implementing and analyzing foundational quantum circuits.
- Designed and evaluated quantum algorithms for data analysis, pattern recognition, and predictive modeling through simulations and visualizations.
- Built projects including a Quantum Stock Predictor, comparing classical and quantum approaches to assess performance and behavior.
- Conducted research on quantum technologies while preparing technical documentation, experiment reports, and project findings.

Scrupulous Enterprises

Software Engineer

Jul 2022 – Jan 2023

- Developed and maintained RESTful backend services using Java and Spring Boot.
- Designed API integrations between backend services and frontend applications.
- Improved code maintainability through modular architecture and refactoring initiatives.
- Optimized internal dashboards resulting in improved business visibility and operational efficiency.

JSpiders Training and Development Center

Java Full Stack Developer

Jan 2023 – Nov 2023

- Completed intensive training in Core Java, SQL, JDBC, Servlets, Hibernate, Spring, HTML, CSS, JavaScript, and Full Stack Development.
- Developed web-based applications by applying object-oriented programming, database management, and backend development concepts.
- Designed and implemented database queries, CRUD operations, and integrated applications with MySQL databases.
- Gained hands-on experience in software development lifecycle, debugging, problem-solving, and project-based learning.

PreciseVision Management Consultancy Services Pvt. Ltd.

Assistant - Income Tax Proof Verification (IBM Payroll Project)

Dec 2023 – Dec 2024

- Audited payroll tax documentation to ensure compliance with statutory regulations.
- Validated employee records using SAP Payroll systems and identified data discrepancies.
- Improved documentation accuracy and record-keeping efficiency by approximately 15%.
- Collaborated with enterprise payroll operations supporting large-scale employee verification processes.

Technical Projects

Online Banking System:

- Architected a secure banking platform with authentication, fund transfers, and transaction tracking.
- Developed backend services using Spring Boot, Hibernate, JPA, and MySQL.
- Implemented secure transaction workflows ensuring data integrity and reliability.
- Built responsive UI components using HTML, CSS, Bootstrap, and jQuery.

AI-Driven Career Growth and Job Assistance Platform:

- Created an AI-based career intelligence platform for salary prediction, job matching, and skill-gap analysis.
- Implemented REST APIs using FastAPI for resume parsing and recommendation workflows.
- Applied NLP and CNN models for salary estimation with normalized salary bands.
- Designed an explainable similarity scoring engine for resume-job matching.
- Tech Stack: Python, FastAPI, React, PostgreSQL, NLP, CNN.

Privacy-Aware Proximity Marketing System:

- Implemented backend APIs using FastAPI to support ML-driven advertisement prediction workflows.
- Developed a React-based admin dashboard for campaign management.
- Built a token-based transaction simulator demonstrating privacy-aware ad exchanges.

QuickBite - Full Stack Food Delivery PWA:

- Developed a Progressive Web App using HTML5, Tailwind CSS, and JavaScript.
- Integrated Firebase and ASP.NET Core Web APIs for authentication and order management.
- Implemented real-time updates to improve customer order visibility.

Depression, Anxiety & Overthinking Detection:

- Designed a biosignal-based classification system for emotion and mental state analysis.
- Applied speech recognition techniques and CNN models achieving approximately 82% accuracy.
- Focused on feature extraction, model evaluation, and ethical AI practices.

Key Technologies

Java | Spring Boot | Hibernate | REST APIs | SQL | MySQL | Microservices | React | FastAPI | Docker | Git | Firebase | Machine Learning | Quantum Machine Learning

Certifications

JP Morgan Software Engineering Job Simulation - Forage

Spring Boot Certification - Scaler

Full Stack Java Developer - JSpiders

Artificial Intelligence and Machine Learning - Internshala